

Theorem.

Every subset of $\mathbb{Z}^+$ is countable.

Let A be a subset of $\mathbb{Z}^+$

Case 1: A is finite $\Rightarrow A$ is countable. (always in dex it)

Case 2: A is infinite.

There is a bijection from A to $\mathbb{Z}^+$.

A is non-empty.

① A has minimum element.
 $a_1 = \min(A)$.

② Consider the new set $A - \{a_1\}$.

Since A is infinite, this new set is also non-empty.

This new set $A - \{a_1\}$ will also have a minimum. $a_2 = \min(A - \{a_1\})$.

③ Continuing this inductive argument
 $a_n = \min(A - \{a_1, a_2, \dots, a_{n-1}\})$.

④ As A is infinite, this process never stops.

⑤ We get an infinite sequence

$$a_1 < a_2 < a_3 \dots$$

⑥ ~~Now~~ Define a function $f: \mathbb{Z}^+ \rightarrow A$
by $f(n) = a_n$.

To prove:
 $f(n)$ is bijective.

⑦

⑧ If $n \neq m$, without any loss of generality $n < m$, by our defⁿ
 $a_n < a_m \Rightarrow f(n) \neq f(m)$

$f(n)$ is injective.

(9)

Let x be any element in A .

Since x is a positive integer there at

~~most~~ most $x-1$ elements in A

that are strictly smaller than x .

Let the ^{no. of} elements smaller than x be k .

~~Then x must be the~~

Then x must be the $(k+1)^{\text{th}}$ smallest element in A . which means

$$f\left(\frac{k+1}{1}\right) = x.$$

Hence f is surjective.

f is bijective.

Corollary: Subset of any countable set is countable.

S is the countable set.
To prove: $A \subseteq S$ is also countable.

Case 1: A is finite.
 $\Rightarrow A$ is countable.

Case 2: A is infinite.

① Since S is countable,
there is bijection from S to $\mathbb{Z}^+$.

② Call this bijective function f

③ Restrict f to a "smaller" set A
creating a new function $g: A \rightarrow \mathbb{Z}^+$
s.t., $g(x) = f(x)$, $\forall x \in A$.

④ Image of $g(A)$: subset of $\mathbb{Z}^+ \rightarrow$ countable.

Theorem:

~~The set $(0, 1] = \{x | x \in \mathbb{R} \text{ and } 0 < x \leq 1\}$~~
The set $(0, 1] = \{x | x \in \mathbb{R} \text{ and } 0 < x \leq 1\}$ is not countable (uncountable).

Proof: If $(0, 1]$ is countable then we can write this set as a sequence of distinct terms
 $(0, 1] = \{r_1, r_2, r_3, \dots\}$.

Nos. like 0.5 $\xrightarrow{\text{always expressed as}}$ 0.4999...

$\therefore$ No element in $(0, 1]$ can be represented by a decimal that terminates.

The expansions of r_1, r_2, r_3 can be written as
 Cantor's diagonalization (Georg Cantor).

$$r_1 = 0. \cancel{a_{11}} a_{12} a_{13} a_{14} \dots$$

$$r_2 = 0. a_{21} \cancel{a_{22}} a_{23} a_{24} \dots$$

$$r_3 = 0. a_{31} a_{32} \cancel{a_{33}} a_{34} \dots$$

$$\vdots$$

$$r_n = 0. a_{n1} a_{n2} a_{n3} a_{n4} \dots$$

$$\vdots$$

$$a_{ij} \in \{0, 1, 2, \dots, 9\}, \quad i, j \in \mathbb{Z}^+$$

Construct a number $r = 0. b_1 b_2 b_3 \dots$
 where for each $k \in \mathbb{Z}^+$

$$b_k = \begin{cases} 3 & \text{if } a_{kk} \neq 3 \\ 7 & \text{if } a_{kk} = 3 \end{cases}$$

$$b_1 \neq a_{11}$$

$$b_2 \neq a_{22}$$

$$b_3 \neq a_{33}$$

$$\vdots$$

$$r \neq r_1 \quad (b_1 \neq a_{11})$$

$$r \neq r_2 \quad (b_2 \neq a_{22})$$

$$r \neq r_3 \quad (b_3 \neq a_{33})$$

$$\vdots$$

$$r \neq r_k$$

The sequence
 does not
 include r
 But $r \in \mathbb{R}$.
 r cannot be
 indexed.

is r a real number?

$$\therefore r \in \mathbb{R}.$$

Hence $[0, 1]$ is
 uncountable $\square$.

Theorem: The set $\mathbb{Z}^+ \times \mathbb{Z}^+$ is Countable?
 Establish a bijection $f: \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

Arrange all pairs $(m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ in a grid.

(1,1)	(1,2)	(1,3)	(1,4)	...
(2,1)	(2,2)	(2,3)	(2,4)	...
(3,1)	(3,2)	(3,3)	(3,4)	...
(4,1)	(4,2)	(4,3)	(4,4)	...
⋮	⋮	⋮	⋮	⋮

$$D_k = \{(m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+, m+n = k\}$$

$$D_2: (1,1) \rightarrow \begin{matrix} m, n \\ 1+1 = 2 \end{matrix}$$

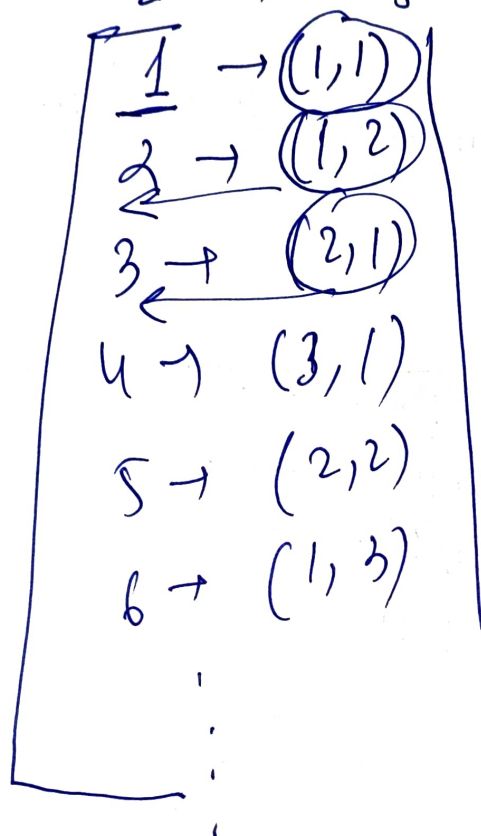
$$D_3: (1,2), (2,1) \rightarrow \begin{matrix} m, n \\ 1+2 \\ 2+1 \end{matrix}$$

$$D_4: (3,1), (2,2), (1,3) \dots$$

Each pair appears exactly once.

Every D_k is finite

The listing $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+ \times \mathbb{Z}^+$
 $\mathbb{Z}^+ \rightarrow$ Diagonal braces.



(m, n) will after $(D_2, D_3, \dots, D_{m+n-1})$
+ m .

$$\text{position of } (m, n): \binom{m+n-1}{2} + m.$$

Theorem: The set $\mathbb{Q} \cap (0, 1]$ is ? Countable?
 $= \{s \mid s \in \mathbb{Q} \text{ and } 0 < s \leq 1\}$

Each s can be written as p/q where $p, q \in \mathbb{Z}^+$
& p & q are co-prime.

$$f: \mathbb{Q} \cap (0, 1] \rightarrow \mathbb{Z}^+ \times \mathbb{Z}^+$$

$$f(p/q) = (p, q)$$

Let U be the range of f .

For $p/q, u/v \in \mathbb{Q} \cap (0, 1]$

$$\Rightarrow \text{if } f(p/q) = f(u/v) \Rightarrow (p, q) = (u, v)$$

$\Rightarrow p = u, q = v$
 $\therefore f$ is injective $\Rightarrow f$ is countable.

□

Theorem. $\mathbb{Q}^+$ is ? countable?

Arrange members of $\mathbb{Q}^+$ in a grid.

Rows $\rightarrow$ numerators p
Columns $\rightarrow$ denominators q .

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
...	...	...	...	...

Listing $1 \rightarrow \frac{1}{1}$
 $2 \rightarrow \frac{1}{2}$
 $3 \rightarrow \frac{2}{1}$
 $\vdots$

(*) The set of all programs that you can write is ? Countable?

Observation:

Every program is finite string of
a fixed & finite alphabet
ASCII char set.
letters, digits, punctuation.

↓
128.

Alphabet : Σ
Strings : Σ^*

$\Sigma = \{a, b\}$
 $\Rightarrow aa, ab, aab,$
 $abba \dots$

Σ^* can be actually enumerated
just like $\mathbb{Z}^+ \times \mathbb{Z}^+$.

Q: Which function generates $\left\{\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{1}{3}, \dots\right\}$

For $n \in \mathbb{Z}^+$, $a_1 = 1$, $\frac{a_{2n} = a_n, a_{2n+1} = a_n + a_{n+1}}{n^{\text{th}} \text{ rational no. } f(n) = \frac{a_n}{a_{n+1}}}$

$$a_1 = 1, a_2 = a_1 = 1$$

$$a_3 = a_1 + a_2 = 1 + 1 = 2$$

$$a_4 = a_2 = 1$$

$$a_5 = a_3 + a_2 = 1 + 2 = 3.$$

Theorem: The union of two countable sets is countable.

$\Rightarrow \mathbb{Z}^+ \cup \mathbb{Z}^- = \mathbb{Z}$ is also countable.

$\Rightarrow \mathbb{Q}^+ \cup \mathbb{Q}^- = \mathbb{Q}$ " " "

Grouping in Σ^*

Group the Diagonals:

- Group the Diagonals.
- (1) ~~First~~ First list all strings of length 0
- (2) Then list all strings of length 1
- (3)
- a u u u u
- a u u u
- a u u
- a u
- a

$$\Sigma = \{a, b\}.$$

Length restricted
lexicographically
sorted.

length 0

length 1

length 2

length 3

lexicographically sorted.

ϵ #1	b #3		
a #2	ab #5	ba #6	bb #7
aa #4	aab #9		
aaa #8			bbb #15

∴ The no. of programs that we can write is countable.

(*) The set of all possible problems is "uncountable".

In reality: There are only a very few problems for which we can write programs.

~~A~~ A problem: Given any string of text as input, answer "yes" or "no" according to some fixed rule.

(1) Formally $f: \Sigma^* \rightarrow \{\text{"yes"}, \text{"no"}\}$.

(2) We say that a program solves a problem if for every input w the program produces a correct answer $f(w)$.

$\Rightarrow$ There exists a problem that
NO PROGRAM SOLVES!!

Observation: f is completely determined by the inputs to which it answers "yes".

$$L_f = \{w \in \Sigma^* : f(w) = \text{yes}\} \subseteq \Sigma^*$$

Collection of all L_f 's.

Our job reduces to show that $P(\Sigma^*)$ is uncountable.

① Σ^* is countably infinite.

We can write the elements of Σ^* as a sequence $s_1, s_2, s_3, \dots$

② By way of contradiction $\rightarrow P(\Sigma^*)$ can also be written as an ordered sequence $L_1, L_2, \dots$

③ Build the table where row i and column j is interpreted as s_j in L_i (Y/N).

④ We construct the diagonal set

$$D = \{ s_i \in \Sigma^*, \underline{s_i \notin L_i} \}$$

⑤ D is a perfectly valid subset of Σ^*

⑥ If our list is complete then D must be equal to some L_k .

⑦ Ask if $\underline{s_k \in D}$.
then by defⁿ D , $s_k \notin L_k$.

But we noted that $D = L_k$

$\therefore \underline{s_k \notin D}$
Contradiction.

⑧ As If $\underline{s_k \notin D}$
Then $s_k \notin L_k$ is false.
 $s_k \in L_k = D \Rightarrow \underline{s_k \in D}$
Contradiction

$\therefore D$ is nowhere to be found in the sequence of L_i .
Let $P(\Sigma^*)$ is uncountable $\square$

	s_1	s_2	s_3	s_4	s_5	...
L_1	$\textcircled{N}$	Y	N	Y	N	...
L_2	Y	$\textcircled{Y}$	N	N	Y	-
L_3	N	Y	$\textcircled{N}$	Y	N	-
L_4	Y	Y	Y	$\textcircled{N}$	N	-
L_5	N	N	Y	Y	$\textcircled{Y}$	-
...	...	...	...	...	...	...
D	Y	N	Y	Y	N	-

$\therefore P(\Sigma^*)$ is uncountable.